1 Introduction

We begin by revisiting a fundamental decision-making problem, and how it is treated within the framework of Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The utility of each action can be quantified by a real number, which depends on the unknown state of the environment. Specifically, let a denote an action, and x represent the unknown environment state. The utility function $U(a, x)$ quantifies the desirability of each action.

Moreover, an evidence e of this unknown environment state exists, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find the optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence.

According to Bayesian decision theory, the optimal strategy is the one that maximizes the expected utility:

$$S_{\text{opt}} = \arg\max_{S} \sum_{a} \sum_{x,e} U(a,x) S(a \mid e) P^*(x,e).$$

Here, $P^*(X, E)$ is the joint probability distribution of the environment state and the evidence. However, a fundamental question arises: How do we obtain this distribution? Is there an objective distribution inherent in the environment that we need to somehow discover?

Answering this question is not straightforward. From a subjective perspective on probability, the answer is negative. There is no such objective distribution, and P^* exists only in our minds. It serves as a tool that helps us regulate our reasoning—an intermediate measure created within the Bayesian decision-making framework to ensure we stay consistent with our assumptions and common sense principles.

Considering the subjective viewpoint, the determination of P^* relies on one's prior knowledge and assumptions. When we have no assumptions and no prior knowledge, the set of candidate probabilities for P^* simply consists of all possible probability measures. As we make assumptions or obtain some prior knowledge, those probability measures that do not align with our assumptions or knowledge are eliminated from this set. Consequently, the set of acceptable probability measures becomes more refined. We refer to this refined set as a probabilistic model:

Definition 1 (Probabilistic Model)**.** We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of acceptable probability measures on $\mathcal{S}$.

One of the most commonly used assumptions of this kind in the statistical modeling is the exchangeability assumption. We say a sequence of random variables $X_1, X_2, \ldots$ is exchangeable if the ordering of variables does not provide any useful information. Or in other words, for an exchangeable sequence of random variables, the joint probability is the same for any permutation of variables. This property holds true for every subset of the sequence.

For example, if each X_i represents a binary variable, the joint probability depends solely on the total count of 1s and 0s, and the specific order in which these variables appear becomes irrelevant.

When a sequence of random variables is exchangeable, de Finetti's theorem states that their joint probability has a special representation:

$$P(x_1, x_2, \ldots) = \int \prod_i \pi(x_i \mid \theta) p(\theta) d\theta.$$

This means, exchangeable observations are conditionally independent relative to some latent parameter θ.

Moreover, the theorem states that if each X_i is discrete, the distribution $\pi(x_i \mid \theta)$ corresponds to the multinomial distribution.

Definition 2 (Parametric Model)**.** Let Θ be a random variable. A *parametric model* is a probabilistic model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p, such that for every $s \in \mathcal{S}$

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \tag{1}$$

while π is a known probability distribution. We refer to π as the *likelihood* distribution.

Now, consider our decision-making problem: Suppose the evidence e consists of a sequence of previous experiences of the environment. If we assume that our environment remains unchanged over time (making this sequence exchangeable), to fully determine P^*, we only need to determine the prior probability of the latent parameter $p(\theta)$.

Clearly, one approach to selecting a suitable prior distribution $p(\theta)$ is to rely on our prior knowledge about the environment. However, what if such knowledge is unavailable?

Even without prior knowledge, it is not difficult to identify some distributions that are not suitable choices for $p(\theta)$. For instance, any distribution that assigns zero probabilities to certain values of θ is not a good choice. If we use such a prior, no evidence can convince us that the value of θ lies within those excluded values.

Unfortunately, arguments like this one can only rule out a small portion of the potential distributions for $p(\theta)$, leaving us with numerous options and an unclear path for selection.

For now, let's assume that we have somehow determined P^*, and shift our focus to finding the optimal strategy.

It can be shown that there always exists a deterministic optimal strategy. That means, we can consider a strategy as a function of the evidence, rather than a conditional distribution:

$$S_{\text{opt}} = \arg\max_S \sum_{x,e} U(S(e), x) P^*(x, e).$$

We can easily verify that the optimal strategy, for each evidence e, chooses the action that maximizes the expected utility conditioned on e:

$$\begin{aligned} S_{\text{opt}}(e) &= \arg\max_a \mathbb{E}_{P^*}\left[U(a, x) \mid e\right] && (2) \\ &= \arg\max_a \sum_x U(a, x) P^*(x \mid e)\cdot && (3) \end{aligned}$$

This implies that the only distribution which affects the optimal strategy is $P^*(X \mid E)$. Note that the value of the expected utility depends on $P^*(E)$, but the actions that construct the optimal strategy are not affected by $P^*(E)$.

Suppose we decide to use a different evidence, denoted as E', instead of E. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can straightforwardly answer this question. To compare any two strategies, we can simply compare their expected utilities. The strategy with the greater expected utility is the better one.

Note that in reality, we cannot freely choose any evidence we want. The availability of evidence depends on the environment. We can only choose evidence that is a function of the evidence provided by the environment, allowing us to determine its value by observing the original evidence.

Utilizing a strategy that uses a simpler evidence, resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for E is always better than every strategy that uses E'. Hence, at first glance, it appears that Bayesian decision theory does not favor feature selection.

From a certain perspective, it could be argued that adopting a strategy that uses simpler evidence leads to ignoring some parts of available information, and therefore can not be justified by Bayesian decision theory. However, in practice, we have to use feature selection because without it the original learning task requires impractical large datasets.

The key point here is that the space of strategies using E' is a subset of strategies that use E. When we are searching for the best strategy for E, we are implicitly evaluating all simpler forms as well. Hence, we believe if we choose an appropriate probability measure P^*, feature selection should be performed automatically when we are finding the optimal Bayesian strategy.

Recall that, in parametric models, the probability measure P^* is determined by the prior distribution of the model parameter. Unfortunately, many commonly used priors in statistics, such as the flat prior, do not exhibit feature selection characteristics.

This seems to come from the fact that in these commonly used probability measures, the estimate for the distribution of $X \mid E'$ differs from what we intuitively estimate. This estimate tends to have a much higher entropy. As a result the optimal strategy for E' is mostly ignored by the probability measure because for any value of the evidence it yields a relatively low expected utility.

It is reasonable to consider that if we find a probability measure which has a closer estimate to our intuitive estimate for $X \mid E'$, then it will show some interesting feature selection characteristics. Throughout this discussion, we will further investigate this idea and try to formulate a systematic approach to determine such probability measures.

2 Central Distribution

Assume that in an environment we are interested in the probability of a random variable X. We have decided on a probabilistic model $\mathcal{M}$ for the environment, and without any particular prior knowledge, we have chosen a distribution P from our model.

We ask an expert—a person familiar with the environment—to evaluate our distribution P. Presumably, instead of P, that expert has his own beliefs about the environment and thinks we should use a different distribution Q. Consequently, when evaluating P, he would try to measure the loss of using the incorrect distribution $P(X)$, while believing that the correct distribution is Q.

Let's assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left(Q_X \parallel P_X\right)$. It seems reasonable as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}\left(Q_X \parallel P_X\right)$ effectively is a measure of the divergence of P from Q, when Q is the believed true distribution.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence. Alternatively, by adopting a more minimal set of properties, we can prove that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also consider KL divergence as the appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose that $\mathbb{D}\left(Q_X \parallel P_X\right)$ is a divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let Ω be a random variable. If $U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have*

$$\mathbb{D}\left(Q_\Omega \parallel U_\Omega\right) = \mathbb{H}\left[U_\Omega\right] - \mathbb{H}\left[Q_\Omega\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two random variables X, Y*

$$\mathbb{D}\left(Q_{X,Y} \parallel P_{X,Y}\right) = \mathbb{D}\left(Q_X \parallel P_X\right) + \sum_x \mathbb{D}\left(Q_{Y|x} \parallel P_{Y|x}\right) Q(x)\cdot$$

If $\mathbb{D}$ satisfies these properties, then it is the Kullback-Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for X. We define a new continuous random variable Ω and let

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega \mid X)$ and $P(\Omega, X)$ are uniform distributions, and $\mathbb{H}\left[P_{\Omega|x}\right] = \ln P(x)$, $\mathbb{H}\left[P_{\Omega,X}\right] = 0$.

We define $Q(\Omega \mid x)$ to be an arbitrary distribution that has the same support as $P(\Omega \mid x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) &= \mathbb{H}\left[P_{\Omega,X}\right] - \mathbb{H}\left[Q_{\Omega,X}\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\omega \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned}\mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) &= \mathbb{H}\left[P_{\Omega|x}\right] - \mathbb{H}\left[Q_{\Omega|x}\right] \\ &= \ln P(x) + \int q(\omega \mid x) \ln q(\omega \mid x) d\omega\cdot\end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned}\mathbb{D}\left(Q_X \parallel P_X\right) &= \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) - \sum_x \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) Q(x) \\ &= \sum_x Q(x) \left(\int q(\omega \mid x) \ln \frac{q(\omega, x)}{q(\omega \mid x)} d\omega - \ln P(x)\right) \\ &= \sum_x Q(x) \left(\ln Q(x) \int q(\omega \mid x) d\omega - \ln P(x)\right) \\ &= \sum_x Q(x) \ln \frac{Q(x)}{P(x)}\cdot \qquad \square\end{aligned}$$

In this theorem, the first property simply states that the entropy of a distribution should be related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X, Y)$ from $Q(X, Y)$ is a measure of the generic loss of using $P(X, Y)$ when we believe $Q(X, Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with probability $Q(x)$, it would also include the loss of using $P(Y \mid x)$. This property indicates that for combining these individual terms we should use addition.

Returning to our earlier discussion, this implies that to evaluate P in regard to some random variable X, our expert should use the KL divergence of $P(X)$ from $Q(X)$.

In the context of Bayesian decision theory, for evaluating a probability measure P, it appears reasonable to focus primarily on the optimal Bayesian strategy that is obtained when using P. We may limit the scope of strategies to those that use an evidence from a specific set of random variables $\mathcal{E}$. Our evaluation should not depend on the utility function, since that would imply that our belief about the environment is a function of our benefit.

From Equation 3, we know that the optimal strategy for evidence E depends only on $P(X \mid E)$. Hence, given an observed value e of the evidence, we may calculate the KL-divergence of $P(X \mid e)$ from $Q(X \mid e)$. Then, to obtain a measure that is not conditioned on a specific value of the evidence, we should calculate the expected value of this KL divergence with respect to Q, which represents our expert's belief about the environment.

This way, we will have a divergence measure for evaluating a distribution. What if we find a probability measure C that has a small divergence from any other distribution in our model? Thus, every expert would agree that our probability measure is a good choice. Interestingly, this relates to the concept of Bayesian convergence, where the posterior probability converges to the same distribution regardless of the prior distribution.

Let's revisit our discussion from the previous section. We saw that in Bayesian framework when we have evidence E', such that $E' = f(E)$, the optimal strategy for E always has a greater expected utility than any strategy that uses E'. We stated that we believe...

Definition 3 (ϵ-convergence)**.** In a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, let X be a discrete random variable, and $\mathcal{E}$ be a countable set of discrete random variables, all defined on $\mathcal{S}$. We refer to X as the *query variable*, and $\mathcal{E}$ as the *evidence set.*

We say $X|\mathcal{E}$ (X conditioned on $\mathcal{E}$) *ϵ-converges by* $P \in \mathcal{P}$, if for any random variable $E \in \mathcal{E}$ and any probability measure $Q \in \mathcal{P}$ we have:

$$\mathbb{E}_Q \left[\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right) \right] < \epsilon,$$

where $\mathbb{E}_Q [\,\cdot\,]$ denotes expectation relative to Q and $\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right)$ is the Kullback-Leibler divergence between $P(X \mid e)$ and $Q(X \mid e)$:

$$\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right) = \sum_x Q(x \mid e) \ln \frac{Q(x \mid e)}{P(x \mid e)}.$$

When $\mathcal{E}$ contains a single random variable E, We refer to E as the *evidence variable.* We also simplify our notation and instead of writing $X|\{E\}$, we use $X|E$.

Definition 4 (Divergence Measure)**.** Let X be a query variable, and $\mathcal{E}$ be an evidence set in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of* $P \in \mathcal{P}$ *for* $X|\mathcal{E}$ as follows:

$$\begin{aligned} \mathbb{D}_{X|\mathcal{E}} [P] &= \max_{Q \in \mathcal{P}} \max_{E \in \mathcal{E}} \mathbb{E}_Q \left[\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right) \right] && (4) \\ &= \max_{Q \in \mathcal{P}} \max_{E \in \mathcal{E}} \sum_{x,e} Q(X = x, E = e) \ln \frac{Q(X = x \mid E = e)}{P(X = x \mid E = e)}. && (5) \end{aligned}$$

Note that we usually simplify our notation and instead of writing the full assignment to a random variable, we only write the assigned value. For example, instead of $Q(X = x, E = e)$ we write: $Q(x, e)$.

Proposition. $X|\mathcal{E}$ ϵ-converges by P, if and only if:

$$\mathbb{D}_{X|\mathcal{E}}\left[P\right] < \epsilon\cdot$$

Definition 5 (Central Distribution)**.** In a model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, we define the *central distribution* for a query variable X, conditioned on the evidence set $\mathcal{E}$, to be a distribution $C_{X|\mathcal{E}} \in \mathcal{P}$ which minimizes the divergence measure for $X|\mathcal{E}$:

$$C_{X|\mathcal{E}} = \arg\min_{P\in\mathcal{P}} \mathbb{D}_{X|\mathcal{E}}\left[P\right]\cdot$$

Definition 6 (Aggregator)**.** Consider two probabilistic models $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$ and $\mathcal{M}^* \equiv (\mathcal{S}^*, \mathcal{P}^*)$. Let X be a query variable and $\mathcal{E} = \{E_1, E_2, \ldots\}$ be an evidence set in $\mathcal{M}$. Also, let E^*, X^*, and I be random variables defined on $\mathcal{S}^*$ in $\mathcal{M}^*$. We refer to X^*, E^* as the *aggregated* query and evidence variables, and I as the *selector* variable.

Suppose that, for a probability measure $P \in \mathcal{P}$, there is a probability measure $P^* \in \mathcal{P}^*$, such that for any index i, and every values x, and e_i there is some x^*, and e^*, where we have

$$P(X = x, E_i = e_i) = P^*(X^* = x^*, E^* = e^* \mid I = i)\cdot$$

In addition, in P^*, X^* and I are conditionally independent given E^*:

$$P^*(X^* \mid E^*, I) = P^*(X^* \mid E^*)\cdot$$

We call P^* an *aggregator of* P *with respect to* $X|\mathcal{E}$.

Definition 7 (Aggregate Model)**.** Consider a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. Let X be a query variable, and $\mathcal{E}$ be an evidence set in $\mathcal{M}$. We say a model $\mathcal{M}^* \equiv (\mathcal{S}^*, \mathcal{P}^*)$ is the *aggregate model* of $\mathcal{M}$ with respect to $X|\mathcal{E}$, If every probability measure in $\mathcal{M}^*$ is the aggregator of a probability measure in $\mathcal{M}$ with respect to $X|\mathcal{E}$.

Definition 8 (Selective Aggregate Model)**.** Let X be a query variable, and $\mathcal{E}$ be an evidence set in $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We say that $\mathcal{M}^*$, an aggregate model of $\mathcal{M}$ with respect to $X|\mathcal{E}$, is *selective*, if for every $P \in \mathcal{P}$ and any value i of the selector variable I, there is an aggregator of P in $\mathcal{M}^*$, such that $P^*(i) = 1$.

Theorem 2. *Let X be a query variable, and $\mathcal{E}$ be an evidence set in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. Suppose that $\mathcal{M}^* \equiv (\mathcal{S}^*, \mathcal{P}^*)$ is a selective aggregate model of $\mathcal{M}$ with respect to $X|\mathcal{E}$. Let X^*, E^* be the aggregated query and evidence variables in $\mathcal{M}^*$. If $P^* \in \mathcal{P}^*$ is an aggregator of the probability measure $P \in \mathcal{P}$, with respect to $X|\mathcal{E}$, then we have*

$$\mathbb{D}_{X|\mathcal{E}}\left[P\right] = \mathbb{D}_{X^*|E^*}\left[P^*\right] \cdot$$

In addition, the central distribution for $X^*|E^*$ is an aggregator of the central distribution for $X|\mathcal{E}$.

Proof. Let

$$R^* = \arg\max_{Q^* \in \mathcal{P}^*} \sum_{x^*, e^*} Q^*(x^*, e^*) \ln \frac{Q^*(x^* \mid e^*)}{P^*(x^* \mid e^*)}.$$

Since $\mathcal{M}^*$ is an aggregate model, R^* must be an aggregator of some probability measure $R \in \mathcal{P}$. It follows that

$$\begin{aligned}
\mathbb{D}_{X^*|E^*}\left[P^*\right] &= \sum_{x^*, e^*} R^*(x^*, e^*) \ln \frac{R^*(x^* \mid e^*)}{P^*(x^* \mid e^*)} \\
&= \sum_{i} R^*(i) \sum_{x, e_i} R(x, e_i) \ln \frac{R(x \mid e_i)}{P(x \mid e_i)} \\
&\leq \max_i \sum_{x, e_i} R(x, e_i) \ln \frac{R(x \mid e_i)}{P(x \mid e_i)} \cdot
\end{aligned}$$

This implies

$$\mathbb{D}_{X^*|E^*}\left[P^*\right] \leq \mathbb{D}_{X|\mathcal{E}}\left[P\right] \cdot \tag{6}$$

Similarly, let

$$(R, k) = \arg\max_{(Q, i)} \sum_{x, e_i} Q(x, e_i) \ln \frac{Q(x \mid e_i)}{P(x \mid e_i)} \cdot$$

$\mathcal{M}^*$ is a selective aggregate model. This means that R must have an aggregator R^* in $\mathcal{M}^*$, which selects k by assigning it a probability of 1: $R^*(k) = 1$.

Therefore,

$$\begin{aligned}
\mathbb{D}_{X|\mathcal{E}}\left[P\right] &= \sum_{x,e_k} R(x,e_k)\ln\frac{R(x \mid e_k)}{P(x \mid e_k)} \\
&= \sum_{x^*,e^*} R^*(x^*,e^* \mid k)\ln\frac{R^*(x^* \mid e^*,k)}{P^*(x^* \mid e^*,k)} \\
&= \sum_{x^*,e^*} R^*(x^*,e^*)\ln\frac{R^*(x^* \mid e^*)}{P^*(x^* \mid e^*)} \\
&\leq \mathbb{D}_{X^*|E^*}\left[P^*\right]\cdot
\end{aligned}$$

By Equation 6, we conclude that

$$\mathbb{D}_{X|\mathcal{E}}\left[P\right] = \mathbb{D}_{X^*|E^*}\left[P^*\right]\cdot$$

To prove the second part of the theorem, let $C_{X^*|E^*}$ be a central distribution for $X^*|E^*$. Since $\mathcal{M}^*$ is an aggregate model for $\mathcal{M}$, $C_{X^*|E^*}$ must be an aggregator for a probability measure $C_{X|\mathcal{E}} \in \mathcal{P}^*$. If we assume, by contradiction, that $C_{X|\mathcal{E}}$ is not a central distribution for $X|\mathcal{E}$ in $\mathcal{M}$, then there must be a probability measure $R \in \mathcal{P}$ such that:

$$\mathbb{D}_{X|\mathcal{E}}\left[R\right] < \mathbb{D}_{X|\mathcal{E}}\left[C_{X|\mathcal{E}}\right] = \mathbb{D}_{X^*|E^*}\left[C^*_{X^*|E^*}\right]\cdot$$

On the other hand, since $\mathcal{M}^*$ is a selective model, R must have an aggregator $R^* \in \mathcal{P}^*$, and we have

$$\mathbb{D}_{X|\mathcal{E}}\left[R\right] = \mathbb{D}_{X^*|E^*}\left[R^*\right] > \mathbb{D}_{X^*|E^*}\left[C^*_{X^*|E^*}\right],$$

which is a contradiction. $\square$

Definition 9 (Conditional Divergence Measure)**.** Assume that $\mathcal{M}_\Theta \equiv (\mathcal{S},\mathcal{P})$ is a parametric model. Let X be a query variable and $\mathcal{E}$ be an evidence set in $\mathcal{M}_\Theta$. We define the *divergence measure of* $P \in \mathcal{P}$ *given* θ *for* $X|\mathcal{E}$ as follows:

$$\begin{aligned}
\mathbb{D}_{X|\mathcal{E}}\left[P \mid \theta\right] &= \max_{E\in\mathcal{E}} \mathbb{E}_\pi\left[\mathbb{D}\left(\pi_{X|e,\theta} \parallel P_{X|e}\right) \mid \theta\right] && (7)\\
&= \max_{E\in\mathcal{E}} \sum_{x,e} \pi(x,e \mid \theta)\ln\frac{\pi\left(x \mid e,\theta\right)}{P\left(x \mid e\right)}\cdot && (8)
\end{aligned}$$

In this definition, the query set $\mathcal{X} = \{(X_1,E_1),(X_2,E_2),\ldots\}$ plays a crucial role. Specifically, $\mathcal{X}$ defines the set of random variables for which we are interested in their distributions. Each X represents a random variable whose distribution is important to us, and each E corresponds to the evidence that is available for that variable.

For a parametric model (as defined in Definition 2), it seems reasonable to define a conditional form of our divergence measure, conditioned on the value of the parameter.

To do so, in Equation **??**, we replace normal expectation with conditional expectation. There will be no longer a need to maximize over different distributions, since in a parametric model the parameter fully determines the distribution of the sample space.

Even if we cannot find a distribution that has a small divergence from every distribution, still using the distribution that minimizes the maximum divergence seems a reasonable choice.

In a parametric model, the maximum of divergence happens for a prior distribution that is concentrated at a single value of the parameter θ. That means, for a parametric model which allows any high peak prior, instead of maximizing over the set of all acceptable distributions, we can simply maximize over the parameter space, using the conditional version of the divergence measure.

Lemma 1. *Assume that $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model. Let X be a query variable, and E be an evidence variable in $\mathcal{M}_\Theta$. For every $P \in \mathcal{P}$ we have*

$$\mathbb{D}_{X|E}\left[P\right] \leq \max_{\theta \in \Theta} \mathbb{D}_{X|E}\left[P \mid \theta\right]$$

Proof. Since the Kullback-Leibler divergence is positive, its expected value is also positive, and for any arbitrary $R \in \mathcal{P}$, and parameter value θ we have

$$\mathbb{E}_\pi\left[\mathbb{D}\left(\pi_{X|e,\theta} \parallel R_{X|e}\right) \,\middle|\, \theta\right] = \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{R\left(x \mid e\right)} \geq 0 \cdot$$

Using this, for every distribution P, we obtain

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \leq \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} \cdot$$

Since this inequality holds for any θ, the maximum of the right-hand side w.r.t. θ is also greater than or equal to the maximum of the left-hand side:

$$\max_{\theta \in \Theta} \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \leq \max_{\theta \in \Theta} \mathbb{D}_{X|E}\left[P \mid \theta\right] \cdot \tag{9}$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x,e) = \int \pi(x,e \mid \theta) r(\theta) d\theta$, where $r(\theta)$ is a prior probability distribution. Trivially, we have

$$\sum_{x,e} \pi(x,e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \leq \max_{\theta \in \Theta} \sum_{x,e} \pi(x,e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \cdot$$

This inequality holds for every value of θ. Therefore, we can multiply this inequality by $r(\theta)$, and integrate. Notice that the right hand side is not a function of θ and it remains unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R \left[\mathbb{D} \left(R_{X|e} \parallel P_{X|e} \right) \right] \leq \max_{\theta \in \Theta} \sum_{x,e} \pi(x,e \mid \theta) \ln \frac{R(x \mid e)}{P(x \mid e)} \cdot$$

Now from Inequality 9, for *any* distribution P and R in $\mathcal{M}_\Theta$, we have

$$\mathbb{E}_R \left[\mathbb{D} \left(R_{X|e} \parallel P_{X|e} \right) \right] \leq \max_{\theta \in \Theta} \mathbb{D}_{X|E} \left[P \mid \theta \right],$$

which proves the lemma. □

Theorem 3. *Suppose $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model. Let X be a query variable, and E be an evidence variable in $\mathcal{M}_\Theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:*

$$\mathbb{D}_{X|E} \left[P \right] = \max_{\theta \in \Theta} \mathbb{D}_{X|E} \left[P \mid \theta \right] \cdot \tag{10}$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let

$$\theta^* = \arg\max_{\theta \in \Theta} \mathbb{D}_{X|E} \left[P \mid \theta \right] \cdot$$

In $\mathcal{M}_\Theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(s \mid \theta^*)$ for every $s \in \mathcal{S}$. Obviously, R and $\pi(\cdot \mid \theta)$ define the same distribution for any random variable, and we have

$$\mathbb{E}_R \left[\mathbb{D} \left(R_{X|e} \parallel P_{X|e} \right) \right] = \mathbb{E}_\pi \left[\mathbb{D} \left(\pi_{X|e,\theta^*} \parallel P_{X|e} \right) \mid \theta^* \right] \cdot \tag{11}$$

On the other hand

$$\max_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right) \right] \geq \mathbb{E}_R \left[\mathbb{D} \left(R_{X|e} \parallel P_{X|e} \right) \right] \cdot$$

Notice that the left hand side is $\mathbb{D}_{X|E} \left[P \right]$, and by Equation 11

$$\mathbb{D}_{X|E} \left[P \right] \geq \max_{\theta \in \Theta} \mathbb{D}_{X|E} \left[P \mid \theta \right] \cdot$$

This inequality and Lemma 1 prove the theorem. □

Proposition. Let X be a query variable, and $\mathcal{E}$ be an evidence set in $\mathcal{M}_\Theta$. If the conditions of Theorem 3 holds, then for any $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{X|\mathcal{E}}[P] = \max_{\theta\in\Theta} \mathbb{D}_{X|\mathcal{E}}[P \mid \theta] \cdot$$

Lemma 2. *Let p, and q be two probability distributions for a random variable Θ. If for any θ, and a constant λ, we have*

$$\ln \frac{p(\theta)}{q(\theta)} = \lambda$$

then $\lambda = 0$.

Proof. Suppose, for the sake of contradiction, that $\lambda > 0$. This implies for every θ we have

$$\frac{p(\theta)}{q(\theta)} > 1 \cdot$$

Since $q(\theta) > 0$, we can conclude that $p(\theta) > q(\theta)$. This inequality holds for every θ, and therefore

$$\int p(\theta)d\theta > \int q(\theta)d\theta,$$

which is a contradiction.

If we assume $\lambda < 0$, similarly we can conclude $\int p(\theta)d\theta < \int q(\theta)d\theta$, which is a contradiction again. Hence, λ cannot be positive or negative and we must have $\lambda = 0$. □

Theorem 4. *Let X be a query variable, and E be an evidence variable in a parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$. Assume that each value of the parameter θ, can be represented as a pair (α, β), and for every α, β, and e we have*

$$p(\beta \mid \alpha, e) = p(\beta \mid \alpha) \cdot \tag{12}$$

If for some probability measure $P \in \mathcal{P}$, the conditional divergence of P is a real valued function of α:

$$\mathbb{D}_{X|E}[P \mid \theta] = \lambda(\alpha),$$

then P is the central distribution for $X|E$, and

$$\lambda(\alpha) = \ln \int \phi(\alpha, \beta)d\beta,$$

where

$$\phi(\alpha, \beta) = \exp\left\{\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{p(\theta \mid x, e)}{p(\alpha \mid e)}\right\} \cdot$$

Proof. By Bayes' theorem and Equation 12 we have

$$\begin{aligned}\frac{\pi(x \mid e, \theta)}{P(x \mid e)} &= \frac{p(\theta \mid x, e)}{p(\beta \mid \alpha, e)p(\alpha \mid e)} \\ &= \frac{p(\theta \mid x, e)}{p(\beta \mid \alpha)p(\alpha \mid e)}.\end{aligned}$$

Now, by simple algebra we can verify that

$$\mathbb{D}_{X|E}\left[P \mid \theta\right] = \ln \frac{\phi(\alpha, \beta)}{p(\beta \mid \alpha)}$$

We let

$$\rho(\beta \mid \alpha) = \frac{\phi(\alpha, \beta)}{\int \phi(\alpha, \beta) d\beta}$$

Therefore,

$$\mathbb{D}_{X|E}\left[P \mid \theta\right] = \lambda(\alpha) + \ln \frac{\rho(\beta \mid \alpha)}{p(\beta \mid \alpha)}$$

where

$$\lambda(\alpha) = \ln \int \phi(\alpha, \beta) d\beta$$

□

Definition 10 (Generic Parametric Model)**.** We say a parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is *generic*, if for any arbitrary probability distribution r, there is a probability measure $R \in \mathcal{P}$, such that for every $s \in \mathcal{S}$

$$R(s) = \int \pi(s \mid \theta) r(\theta) d\theta \cdot$$

Proposition. A generic parametric model satisfies the conditions of Theorem 3.

Theorem 5 (Minimax theorem [2])**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- *$f(\cdot, y) : \mathcal{X} \to \mathbb{R}$ is concave for fixed y, and*
- *$f(x, \cdot) : \mathcal{Y} \to \mathbb{R}$ is convex for fixed x.*

Then we have that

$$\max_{x \in \mathcal{X}} \min_{y \in \mathcal{Y}} f(x, y) = \min_{y \in \mathcal{Y}} \max_{x \in \mathcal{X}} f(x, y) \cdot \tag{13}$$

Theorem 6. *Consider a generic parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, and let X be a query variable, and E be an evidence variable in $\mathcal{M}_\Theta$. Suppose that for every probability measures $Q_1, Q_2 \in \mathcal{P}$, and any real number $0 \leq \alpha \leq 1$, there is a $Q^* \in \mathcal{P}$, such that for every x, e we have*

$$Q^*(x \mid e) = \alpha Q_1(x \mid e) + (1-\alpha) Q_2(x \mid e)\cdot \tag{14}$$

In this model, the central distribution for $X|E$ can be obtained by:

$$C_{X|E} = \arg\max_{P \in \mathcal{P}} \int \mathbb{D}_{X|E}\left[P \mid \theta\right] p(\theta) d\theta \tag{15}$$

$$= \arg\max_{P \in \mathcal{P}} \int \sum_{x,e} p(x,e,\theta) \ln \frac{\pi(x \mid e, \theta)}{P(x \mid e)} d\theta, \tag{16}$$

where $P(s) = \int \pi(s \mid \theta) p(\theta) d\theta$, for every $s \in \mathcal{S}$.

Proof. We prove this theorem using the Minimax theorem. For that, we define a functional as follows:

$$f(r, P_{X|E}) = \int \mathbb{E}_\pi \left[\mathbb{D}\left(\pi_{X|e,\theta} \parallel P_{X|e}\right) \mid \theta\right] r(\theta) d\theta,$$

where $r(\theta)$ is an arbitrary probability distribution, and $P_{X|E}$ represents different distributions of X given different values of E, induced by a probability measure $P \in \mathcal{P}$.

Since $r(\theta)$ can be any distribution, maximizing with respect to r is similar to maximizing with respect to θ, and we have

$$C_{X|E} = \arg\min_{P \in \mathcal{P}} \max_r f(r, P_{X|E})\cdot$$

The domain of r is the set of all probability distributions, which is a compact convex set. Moreover, from Equation 14, it follows that the domain of $P_{X|E}$, which is a function of both X and E, is also a compact convex set. Thus, the domain of the functional $f(r, P_{X|E})$ is a compact convex set.

For a fixed $P_{X|E}$, $f(\cdot, P_{X|E})$ is a linear function, which is both concave and convex.

On the other hand, for a fixed r, if we let $R(x,e) = \int \pi(x, e \mid \theta) r(\theta) d\theta$ we have

$$f(r, \cdot) = c - \sum_{x,e} R(x,e) \ln P(x \mid e), \tag{17}$$

where c is not a function of $P_{X|E}$. If we calculate the Hessian matrix of $f(r, \cdot)$, with respect to $P_{X|E}$, we observe that the matrix is diagonal, and each element of the main diagonal is $\frac{R(x|e)}{P(x|e)^2}$. Thus, the hessian matrix of $f(r, \cdot)$ is positive semi-definite and $f(r, \cdot)$ is a convex function. Consequently, by Theorem 5

$$\min_{P \in \mathcal{P}} \max_{r} f(r, P_{X|E}) = \max_{r} \min_{P \in \mathcal{P}} f(r, P_{X|E}) \cdot \tag{18}$$

From Equation 17 we can see that for an arbitrary fixed r we have

$$\min_{P \in \mathcal{P}} f(r, P_{X|E}) = \max_{P \in \mathcal{P}} \sum_{x,e} R(x, e) \ln P(x \mid e) \cdot$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative:

$$\mathbb{E}_R \left[\mathbb{D} \left(R_{X|e} \parallel P_{X|e} \right) \right] \geq 0 \cdot$$

By simple algebra we get

$$\sum_{x,e} R(x, e) \ln P(x \mid e) \leq \sum_{x,e} R(x, e) \ln R(x \mid e) \cdot$$

Recall that $\mathcal{M}_\Theta$ is a generic parametric model. That implies $R \in \mathcal{P}$, and we can reach this upper bound by letting $P = R$:

$$\min_{P \in \mathcal{P}} f(r, P_{X|E}) = \int \sum_{x,e} r(x, e, \theta) \ln \frac{\pi(x \mid e, \theta)}{R(x \mid e)} d\theta \cdot$$

By Equation 18, we conclude

$$C_{X|E} = \arg \max_{R \in \mathcal{P}} \int \mathbb{D}_{X|E} \left[R \mid \theta \right] r(\theta) d\theta \cdot \qquad \square$$

Theorem 7. *Under the assumptions of Theorem 6, the central distribution for $X|E$ satisfies:*

$$\mathbb{D}_{X|E} \left[C_{X|E} \mid \theta \right] = \lambda, \tag{19}$$

where λ is a constant.

Proof. By Theorem 6, the central distribution for $X|E$ maximizes

$$\mathcal{F} \left[p(\theta) \right] = \int \sum_{x,e} p(x, e, \theta) \ln \frac{\pi(x \mid e, \theta)}{P(x \mid e)} d\theta,$$

with respect to different choices of the prior distribution $p(\theta)$, subject to the constraint $\int p(\theta) d\theta = 1$.

Applying the Lagrange multiplier technique, the local maximum of $\mathcal{F}$ is a stationary point of the Lagrangian functional:

$$\mathcal{L}\left[p(\theta),\lambda\right]=\int\sum_{x,e}p(x,e,\theta)\ln\frac{\pi(x\mid e,\theta)}{P(x\mid e)}d\theta-\lambda\left(\int p(\theta)d\theta-1\right)\cdot$$

Let us denote the prior distribution corresponding to the central distribution $C_{X|E}$ by $c(\theta)$. Let $p(\theta)=c(\theta)+\epsilon\eta(\theta)$ be a variation of $c(\theta)$, where $\eta(\theta)$ is a continuously differentiable function with $\eta(0)=\eta(1)=0$, and ϵ is a small constant.

Since $c(\theta)$ is an extremizing function for $\mathcal{F}\left[p(\theta)\right]$, the lagrangian $\mathcal{L}\left[\epsilon,\lambda\right]$ has a stationary point where $\epsilon=0$, and we have

$$\left.\frac{\partial\mathcal{L}\left[\epsilon,\lambda\right]}{\partial\epsilon}\right|_{\epsilon=0}=0\cdot$$

Note that

$$\begin{aligned}\frac{dp(\theta)}{d\epsilon}&=\eta(\theta),\\ \frac{dP(x,e)}{d\epsilon}&=\int\pi(x,e\mid\theta)\eta(\theta)d\theta,\\ \frac{dP(e)}{d\epsilon}&=\int\pi(e\mid\theta)\eta(\theta)d\theta\cdot\end{aligned}$$

Using the chain rule of calculus, we calculate the partial derivative of $\mathcal{L}\left[\epsilon,\lambda\right]$ relative to ϵ:

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial\epsilon}\ &=\int\left(\sum_{x,e}\pi(x,e\mid\theta)\ln\frac{\pi\left(x\mid e,\theta\right)}{P\left(x\mid e\right)}\right)\eta(\theta)d\theta\\ &\quad-\int\left(\sum_{x,e}\pi(x,e\mid\theta)\frac{\int\pi(x,e\mid\theta)\eta(\theta)d\theta}{P(x,e)}\right)p(\theta)d\theta\\ &\quad+\int\left(\sum_{e}\pi(e\mid\theta)\frac{\int\pi(e\mid\theta)\eta(\theta)d\theta}{P(e)}\right)p(\theta)d\theta\\ &\quad-\int\lambda\eta(\theta)d\theta\end{aligned}$$

Since $P(x,e)=\int\pi(x,e\mid\theta)p(\theta)d\theta$ and $P(e)=\int\pi(e\mid\theta)p(\theta)d\theta$, we get

$$\frac{\partial\mathcal{L}}{\partial\epsilon}=\int\left(-\lambda+\sum_{x,e}\pi(x,e\mid\theta)\ln\frac{\pi\left(x\mid e,\theta\right)}{P\left(x\mid e\right)}\right)\eta(\theta)d\theta\cdot$$

We let $\frac{\partial \mathcal{L}}{\partial \epsilon}\Big|_{\epsilon=0} = 0$. Since $\eta(\theta)$ is an arbitrary function, by applying the fundamental lemma of the calculus of variations and letting $\epsilon = 0$, for every value of θ we have

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{C_{X|E}\left(x \mid e\right)} = \lambda. \qquad \square$$

Consider a scenario where we are trying to predict the outcome of a four-sided dice. This dice is unfair, with each face marked by a binary number: 00, 01, 10, and 11. Before making our prediction, we can roll the dice for several times and use the results to learn the associated probabilities. We assume that the dice rolls are independent.

Let's focus on determining the central distribution for this problem. The first step in this process is to establish our query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$.

In a prediction problem, we are interested in the distribution of the random variable we want to predict, conditioned on the observed data. For example, if we roll the dice for three times before making our prediction, we will need $P(X_4 \mid X_1, X_2, X_3)$, where X denotes the outcome of the i-th dice roll.

Thus, one natural way for defining a query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ is to let each X be the outcome of the i-th dice roll, and define each E as a vector of the previous dice outcomes $(X_1, X_2, \ldots, X_{i-1})$.

Now let us calculate the conditional divergence measure for $\mathcal{X}$ as defined in Definition 9:

$$\begin{aligned}
\mathbb{D}_{X|E}\left[P \mid \theta\right] = &\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_1 \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_2 x_1 \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_3 x_1, x_2 \\
&+ \cdots \\
&+ \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_n x_1, \ldots, x_{n-1}.
\end{aligned}$$

After factoring out $\pi(x_1, \ldots, x_n \mid \theta)$ from every term, we use the chain rule for conditional probability:

$$\mathbb{D}_{X|E}\left[P \mid \theta\right] = \sum_{x_1,\ldots,x_n} \pi(x_1,\ldots,x_n \mid \theta) \ln \frac{\pi(x_1 \mid \theta)\pi(x_2 \mid x_1,\theta)\cdots\pi(x_n \mid x_1,\ldots,x_{n-1},\theta)}{P(x_1)P(x_2 \mid x_1)\cdots P(x_n \mid x_1,\ldots,x_{n-1})}$$
$$= \sum_{x,e} \pi(x,e \mid \theta) \ln \frac{\pi\left(x \mid e,\theta\right)}{P\left(x \mid e\right)} x_1,\ldots,x_n\cdot$$

By Equation 16, for the central distribution we have

$$C_{X|\mathcal{E}} = \arg\max_P \int \left(\sum_{x,e} \pi(x,e \mid \theta) \ln \frac{\pi\left(x \mid e,\theta\right)}{P\left(x \mid e\right)} x_1,\ldots,x_n \right) p(\theta) d\theta$$
$$= \arg\max_P \int \sum_{x_1,\ldots,x_n} p(x_1,\ldots,x_n,\theta) \ln \frac{p(\theta \mid x_1,\ldots,x_n)}{p(\theta)} d\theta\cdot$$

That implies, the central distribution for our problem is the same as the distribution we obtain when using the reference prior [1] for our main parameter.

Imagine playing a game of chance involving our four-sided dice. After rolling the dice, we input the outcome into a mysterious machine. The machine then announces whether we have won or lost the game. We have no clue how the machine works. We are not even sure if the dice outcome truly affects the machine's output. All we know is that the machine is memory-less, and the result of each game is independent.

Our goal is to learn how the machine operates through experimentation. We aim to be able to predict whether we will win the game after observing the dice outcome.

We assume that, first, we roll the dice n times, and record each outcome. Next, we input some of these dice outcomes into the machine and observe the result of each game. As we will see, this assumption is important for deriving the central distribution more easily.

Let us define a query set for this problem. We denote this set as $\mathcal{X} = \{(X_1,E_1),(X_2,E_2),\ldots\}$, where each X is the result of the i-th game, and the evidence E is a vector

$$(X_1,X_2,\ldots,X_{i-1},Y_1,Y_2,\ldots,Y_n),$$

where each Y denotes the i-th dice outcome.

The conditional divergence for $\mathcal{X}$ can be obtained by

$$
\begin{aligned}
\mathbb{D}_{X|E}\left[P \mid \theta\right] = & \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_1 y_1, \ldots, y_n \\
& + \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_2 x_1, y_1, \ldots, y_n \\
& + \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_3 x_1, x_2, y_1, \ldots, y_n \\
& + \cdots
\end{aligned}
$$

After using the chain rule for conditional probability the right hand side simplifies to

$$
\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{P\left(x \mid e\right)} x_1, \ldots, x_n y_1, \ldots, y_n.
$$

Consequently, an alternative way for defining our query set is to define it as a single pair $\{(\mathbf{X}, \mathbf{E})\}$, where $\mathbf{X}$ is the vector of n game's result, and $\mathbf{E}$ is the vector of n dice outcome.

Using the new definition for $\mathcal{X}$, by Equation 19 we have:

$$
\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{C_{X|\mathcal{E}}\left(x \mid e\right)} \mathbf{xe} = \lambda
$$

derive a $\theta = (\alpha, \beta)$ closed form for central dist...

$$
\frac{\pi(\mathbf{x} \mid \mathbf{e}, \theta)}{C_{X|\mathcal{E}}(\mathbf{x} \mid \mathbf{e})} = \frac{\pi(\theta \mid \mathbf{x}, \mathbf{e})}{C_{X|\mathcal{E}}(\theta \mid \mathbf{e})}
$$

sdfd

$$
\sum_{\mathbf{x},\mathbf{e}} \pi(\mathbf{x}, \mathbf{e} \mid \alpha, \beta) \ln \frac{c_{\mathcal{X}}(\beta \mid \mathbf{x}, \mathbf{e})}{c_{\mathcal{X}}(\beta \mid \alpha)}
$$

Assume that we decide to simplify our learning task. We choose to consider only the least significant digit of the dice outcome and ignore the other digit. For example, if the dice outcome is 10, we treat it as 0. This approach decreases the number of possible outcomes of the dice, making the learning process simpler. However, it also reduces our ability to predict the game's result accurately.

In order to find a central distribution for this version of our game, we should slightly modify the definition of the query set. While each X remains unchanged, we need to represent each E as a vector

$$(X_1, X_2, \ldots, X_{i-1}, Y'_1, Y'_2, \ldots, Y'_n),$$

where each Y' represents the least significant digit of the dice outcome. We denote this new query set as $\mathcal{X}'$ and the previous query set as $\mathcal{X}$.

Following similar steps as we did for $\mathcal{X}$, we can show that $\mathcal{X}'$ also has a more compact counterpart, which includes a single pair $(\mathbf{X}, \mathbf{E}')$, where $\mathbf{X}$ is the vector $(X_1, X_2, \ldots, X_n)$, and $\mathbf{E}'$ is the vector $(Y'_1, Y'_2, \ldots, Y'_n)$.

explain how the central distribution for this version is different...

When we have a lot of learning data, there is no point in using this simpler version of our learning task. However, when the available data is not enough for the more complex version, the simpler version can still produce relatively meaningful results, while the data is too small for the more complex version.

Is it possible to have a probabilistic inference framework that is able to automatically choose between these two versions of the problem based on the quality and amount of available data? It seems that the central distributing has this ability. For that, we need to use the central distribution for the query set $\mathcal{X} \cup \mathcal{X}'$. This seems reasonable, since when we want our probabilistic inference framework to choose between two versions of our problem, our query set should include random variables of both versions.

Theorem 8. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{X_1, X_2, X_3, \ldots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x \mid \theta) = \pi(x \mid \omega_i) \cdot \tag{20}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c(\theta) \propto \left(\prod_i c_{\mathcal{X}} \theta \mid \omega_i \rho(\beta \mid \alpha) \right)^{\frac{1}{|\mathcal{X}|}}, \tag{21}$$

where

$$\rho(\beta \mid \alpha) \propto \exp\left\{ \sum_x \pi(x \mid \omega_i) \ln c_{\mathcal{X}} \omega_i \mid x \right\} \cdot$$

Proof. First, we show that the set of distributions of each X is a convex set. Let $Q^*(x) = \alpha Q_1(x) + (1-\alpha) Q_2(x)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \leq \alpha \leq 1$ and arbitrary i, x.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned} Q^*(x) &= \alpha \int \pi(x \mid \theta) q_1(\theta) d\theta + (1-\alpha) \int \pi(x \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(x \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta \cdot \end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha) q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 7 the central distribution of $\mathcal{X}$ satisfies

$$\sum_i \sum_x \pi(x \mid \theta) \ln \frac{\pi(x \mid \theta)}{C_{X|\mathcal{E}}(x)} = \lambda \cdot \tag{22}$$

Moreover, from Equation 23 and Bayes' theorem it follows

$$\begin{aligned} \sum_i \sum_x \pi(x \mid \theta) \ln \frac{\pi(x \mid \theta)}{C_{X|\mathcal{E}}(x)} &= \sum_i \sum_x \pi(x \mid \omega_i) \ln \frac{\pi(x \mid \omega_i)}{C_{X|\mathcal{E}}(x)} \\ &= \sum_i \sum_x \pi(x \mid \omega_i) \ln \frac{c_{\mathcal{X}} \omega_i \mid x}{c(\omega_i)} \\ &= \sum_i \ln \frac{\rho(\beta \mid \alpha)}{c(\omega_i)} \cdot \end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_{\mathcal{X}}\theta, \omega_i = c(\theta)$. Therefore,

$$c(\omega_i) = \frac{c(\theta)}{c_{\mathcal{X}}\theta \mid \omega_i} \cdot$$

Now we rewrite Equation 25:

$$\begin{aligned} |\mathcal{X}| \ln c(\theta) &= \sum_i \ln \left(c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta \mid \alpha) \right) - \lambda \\ c(\theta) &= \left(e^{-\lambda} \prod_i c_{\mathcal{X}}\theta \mid \omega_i \rho(\beta \mid \alpha) \right)^{\frac{1}{|\mathcal{X}|}} \end{aligned}$$

□

Theorem 9. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a set of random variables in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X, so that for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x \mid \theta) = \pi\left(x \mid \omega_i\right) \cdot \tag{23}$$

The prior distribution corresponding to the central distribution for $\mathcal{X}$ in $\mathcal{M}_\Theta$ can be characterized by the following fixed point equation:

$$c_{\mathcal{X}}(\beta \mid \alpha) \propto \left(\prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta)\rho(\beta \mid \alpha)}{c_{\mathcal{X}}(\alpha \mid \alpha)}\right)^{\frac{1}{|\mathcal{X}|}}, \tag{24}$$

where

$$\rho(\beta \mid \alpha) \propto \exp\left\{\sum_{x,e} \pi(x, e \mid \alpha, \beta) \ln c_{\mathcal{X}}(\beta \mid x, e)\right\}\cdot$$

Proof. First, we show that the set of distributions of each X is a convex set. Let $Q^*(x \mid e) = \alpha Q_1(x \mid e) + (1-\alpha)Q_2(x \mid e)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \leq \alpha \leq 1$ and arbitrary i, x, e.

$\mathcal{M}_\Theta$ is a parametric model. Therefore,

$$\begin{aligned}
Q^*(x \mid e) &= \alpha \int \pi(x \mid e, \beta) q_1(\beta \mid e) d\beta + (1-\alpha) \int \pi(x \mid \theta) q_2(\theta) d\theta \\
&= \int \pi(x \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta\cdot
\end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha)q_2(\theta)$ is a valid prior distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$.

Thus, by Theorem 7 the central distribution of $\mathcal{X}$ satisfies

$$\sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{C_{X|\mathcal{E}}\left(x \mid e\right)} = \lambda\cdot \tag{25}$$

Moreover, from Equation 23 and Bayes' theorem it follows

$$\begin{aligned}
\lambda &= \sum_{x,e} \pi(x, e \mid \theta) \ln \frac{\pi\left(x \mid e, \theta\right)}{C_{X|\mathcal{E}}\left(x \mid e\right)} \\
&= \sum_{x,e} \pi(x, e \mid \alpha, \beta) \ln \frac{\pi(x \mid e, \beta)}{C_{X|\mathcal{E}}(x \mid e)} \\
&= \sum_{x,e} \pi(x, e \mid \alpha, \beta) \ln \frac{c_{\mathcal{X}}(\beta \mid x, e)}{c_{\mathcal{X}}(\beta \mid e)} \\
&= \sum_i \ln \frac{\rho(\beta \mid \alpha)}{\beta \mid \alpha}\cdot
\end{aligned}$$

Since each Ω_i is a function of Θ, for $\Omega_i(\theta) = \omega_i$ we have $c_{\mathcal{X}}\theta, \alpha, \beta = c(\theta)$ $c_{\mathcal{X}}\alpha, \alpha = c_{\mathcal{X}}\alpha$. Therefore,

$$\beta \mid \alpha = \frac{c_{\mathcal{X}}(\alpha \mid \alpha)}{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta)} c_{\mathcal{X}}(\beta \mid \alpha)\cdot$$

Now we rewrite Equation 25:

$$|\mathcal{X}| \ln c_{\mathcal{X}}(\beta \mid \alpha) = -\lambda + \sum_i \ln \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta)\rho(\beta \mid \alpha)}{c_{\mathcal{X}}(\alpha \mid \alpha)}$$

Then:

$$\begin{aligned}
c_{\mathcal{X}}(\beta \mid \alpha) &= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\alpha, \beta \mid \alpha, \beta)\rho(\beta \mid \alpha)}{c_{\mathcal{X}}(\alpha \mid \alpha)} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= \left(e^{-\lambda} \prod_i \frac{c_{\mathcal{X}}(\beta \mid \alpha)\rho(\beta \mid \alpha)}{\beta \mid \alpha} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(\prod_i \frac{\rho(\beta \mid \alpha)}{\beta \mid \alpha} \right)^{\frac{1}{|\mathcal{X}|}} \\
&= c_{\mathcal{X}}(\beta \mid \alpha) \left(e^{-\lambda} \prod_i \frac{\rho(\alpha, \beta) c_{\mathcal{X}}(\alpha)}{c_{\mathcal{X}}(\alpha, \beta)\rho(\alpha)} \right)^{\frac{1}{|\mathcal{X}|}} \qquad \square
\end{aligned}$$

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